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ECE 306 Power Electronics

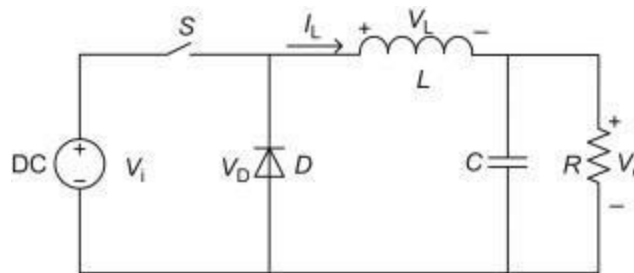
Buck, Boost, Buck-Boost Assignment

BUCK CONVERTER

Why the name “buck converter”?

One meaning of "buck" is to oppose, as in to buck a trend or to buck traffic. Therefore, a buck converter would be one that opposed or did the opposite of a boost converter. The inductor bucks or opposes the voltage when the switch is on. It is also known as a “step-down” converter.

How the buck converter works



In the buck converter, the inductor plays a major role to lower the input voltage. New developments in winding and core technologies are making it advantageous to specify buck and boost inductors at higher power levels, higher frequencies and higher ripple current.

There are two states in the operation process of buck converter: the on-state and off-state of the switch S . The switching element, often a power MOSFET or IGBT, is ideally either opened or closed – off or on – so the buck converter circuit will alternate between two different states.

When the switch is closed (i.e. MOSFET on-state), the sudden rush of current creates a potential across the inductor that opposes the supply voltage, effectively reducing the output voltage. Energy is stored in both the inductor and the capacitor in this state; no current flows through the diode as it is reverse biased.

When the switch is opened the supply current to the inductor is suddenly interrupted. The diode now conducts, providing a return path for the inductor current. This rapid change in current flow reverses the voltage drop across the energized inductor, which then becomes the primary source of output power during this part of the switching cycle.

Building the circuit on Multisim

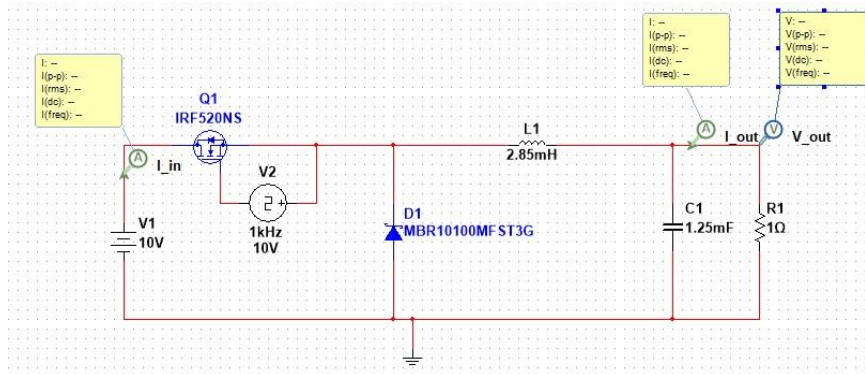


Figure 1 the circuit before "run"

Data for simulation: (these are the values given)

Circuit components: already

MOSFET

Diode

(we want to half the voltage)

Input voltage=10 V

Inductor- 2.85 mH

Output voltage= 5 V

DC Power supply- 10 V

Output current= 5 A

Capacitor- 1.25 mF

Ripple voltage in V= 1%

Resistance- 1 Ohm

Ripple current in I= 10%

Clock source to give the signal to the

Frequency= 1 kHz

higher than calculated 'duty cycle', to match

little bit due to resistance of inductor)

MOSFET (we put 'duty cycle' a little bit

our output to the given value. We do this

because output voltage & current decreases

D=55.7%

$V_{out} V_{in}$

Duty cycle: D=

$$D = \frac{5}{10} = 50\%$$

Inductor value:

$$L = \frac{V_{in} - V_{out}}{2 \Delta i_L} DT$$

Δi_L = Ripple current value in current = 10% of 5A = 0.5

$$T = 1/f = 1/1\text{kHz} = 0.001$$

$$L = \frac{10-5}{2*0.5} * 0.537 * 0.001 = 2.5 \text{ mH}$$

Capacitor value:

$$C = \frac{\Delta i_L}{8 * \Delta V T}$$

ΔV = ripple voltage in output voltage = 1% of 5V = 0.05 V

$$C = \frac{0.5}{8 * 0.05} * 0.001 = 1.25\text{mF}$$

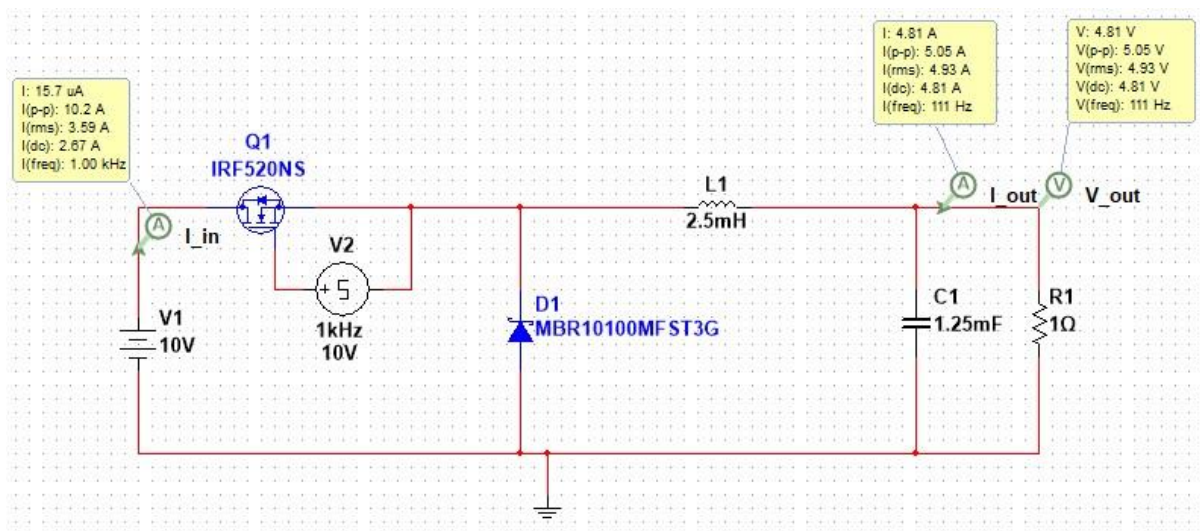


Figure 2 After running circuit in Interactive mode

$$I_{out} = 4.81 \text{ A}$$

$$V_{out} = 4.81 \text{ V}$$

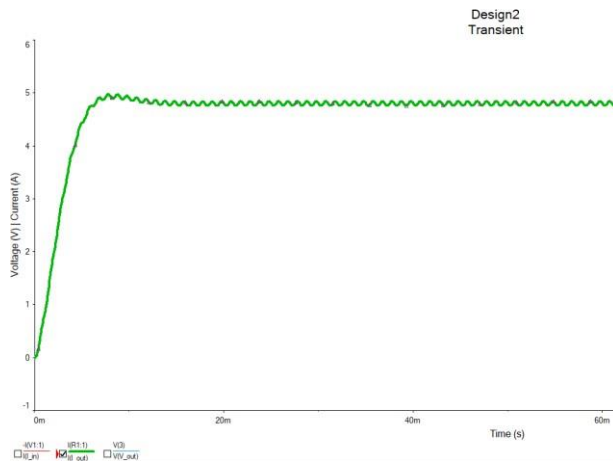


Figure 3 Output Current

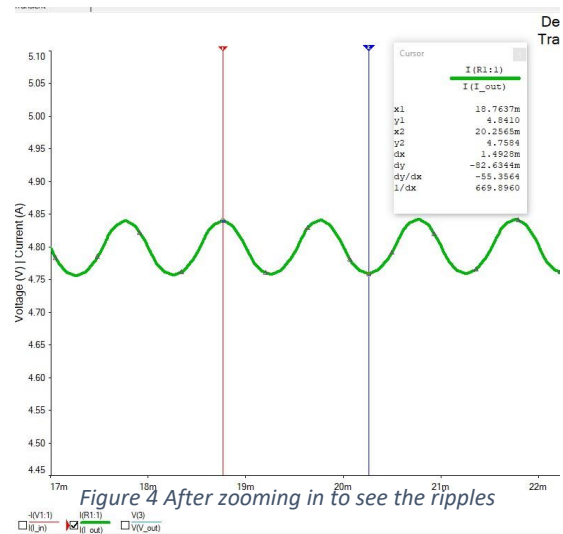


Figure 4 After zooming in to see the ripples

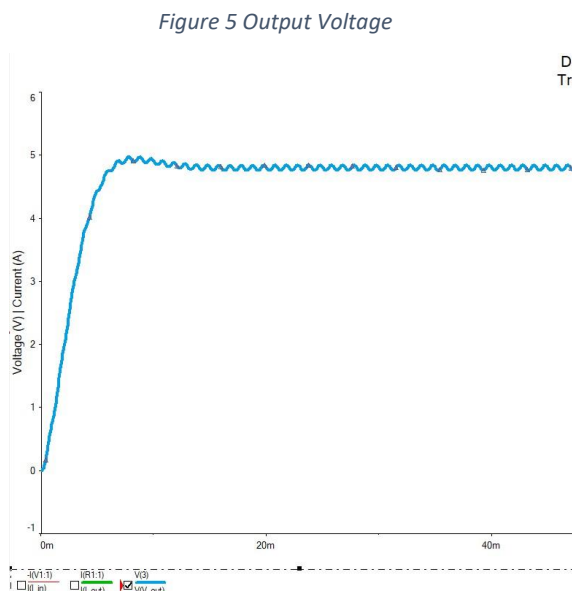


Figure 5 Output Voltage

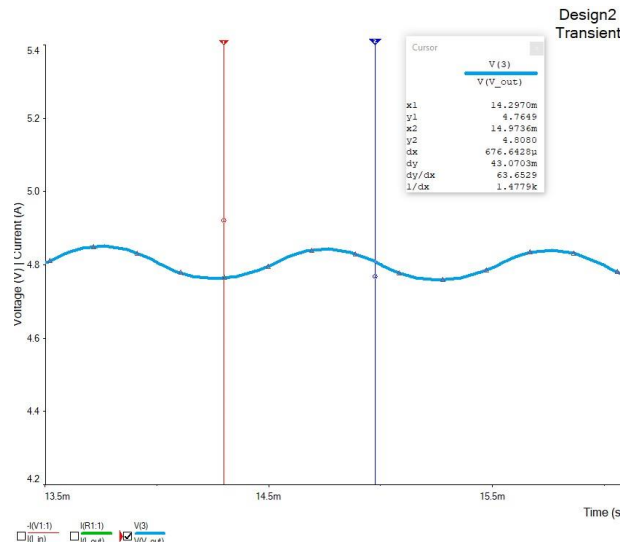


Figure 6 Output Voltage Ripples

Buck converter applications



Figure 7 A PCB motherboard that uses a POL converter

Some common applications of buck converters are: USB on the go, point of load converters for PCs and laptops, Battery Chargers, Quad Copters, Solar Chargers, and power audio amplifiers. These converters are designed to have efficiency of 90% or higher, resulting in low power loss.

Buck Converters in power audio amplifiers

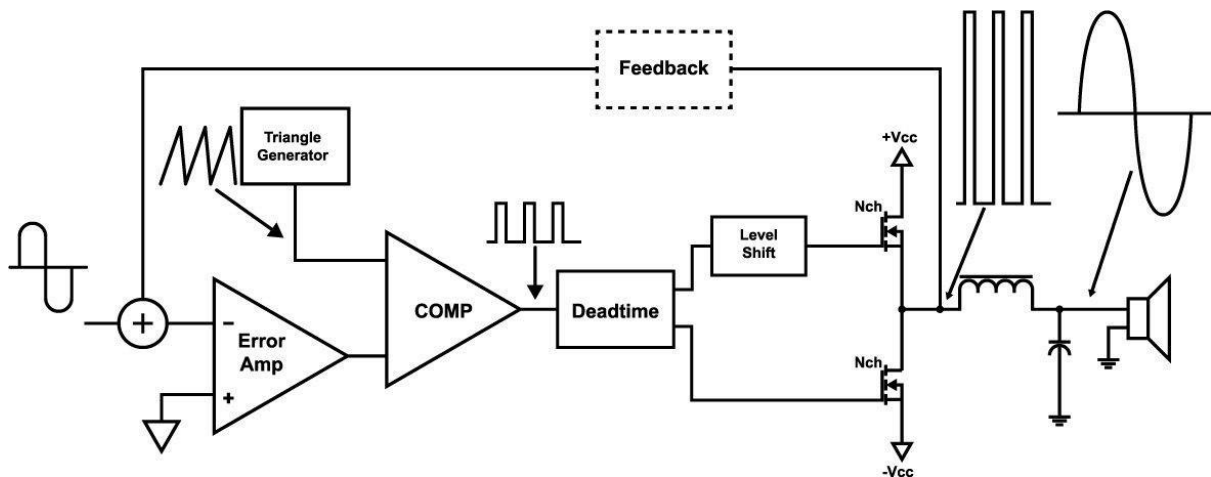


Figure 8 A simplified schematic of a class D amplifier

Class D power audio amplifiers power stage is a synchronous buck converter. One big difference between a DC-DC converter and a class D audio amplifier is that while the DC-DC converter produces a fixed DC voltage over line and load, the class D amplifier intentionally produces an AC signal. The difference is in the feedback loop. Notice that the output stage is a half-bridge followed by an LC filter. A good example of a class D amplifier is NXP's TDA8954.



Figure 9: A quadcopter from Thrust-UAV.

Buck converters are also used in quadcopters

Quadcopters use brushless motors to fly because of their high efficiency and lightweight. Three half-bridges are used to drive the inductive coils of a brushless motor. See figure 9, a simplified schematic of a brushless motor driver. It looks like 3 synchronous buck converters without filter capacitors.

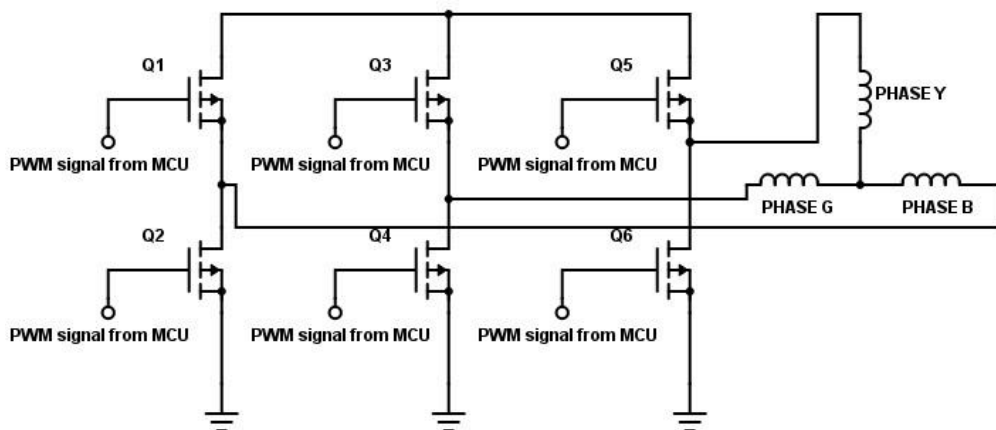


Figure 10 A simplified diagram of a brushless motor controller

BOOST CONVERTER

How the boost converter works

A boost converter is one of the simplest types of switch mode converter. As the name suggests, **it takes an input voltage and boosts or increases it**. Opposite to the buck converter, it is also known as a “step-up” converter.

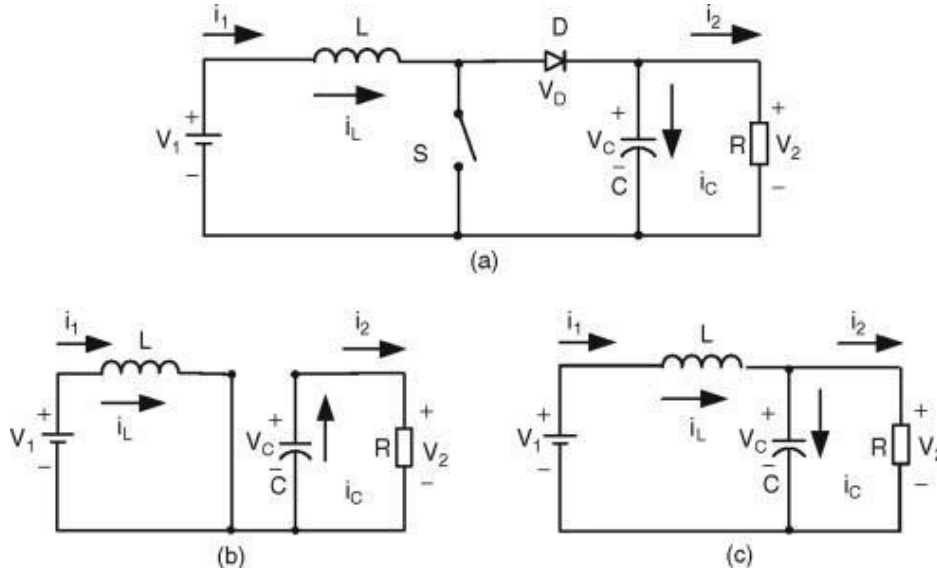


Figure 11 Boost converter: (a) circuit diagram; (b) switch-on equivalent circuit; and (c) switch-off equivalent circuit

All it consists of is an inductor, a semiconductor switch (MOSFET), a diode, a capacitor and a source of a periodic square wave.

There are 3 steps to the boost converter working:

1. Output capacitor is charged to the input voltage minus one diode drop. Nothing happens
2. We turn the switch on, signal source goes high turning on the MOSFET. The output capacitor stays charged, it can't discharge because the diode is back-biased.

The power source doesn't short circuit immediately (because the inductor makes the current slow). A magnetic field builds up around the inductor.

3. The MOSFET is now OFF and the current to the inductor is interrupted.

By its nature, since the inductor likes to maintain smooth current, it doesn't like sudden current change. When this happens, it generates voltage with opposite polarity to the voltage that was originally supplying it (by using energy stored in magnetic field).

We notice that the inductor now acts like a voltage source in series with the supply voltage. The diode is now forward biased.

The output capacitor is now charged to a higher voltage than before, which means the boost converter has accomplished dc-dc conversion.

Building the circuit on Multisim

Data for simulation:

(we want to double the voltage)

Input voltage=10 V

Output voltage= 20 V

Output current= 5 A

Ripple voltage in V= 1%

Ripple current in I= 10%

Frequency= 1 kHz

Duty cycle:

$$D = (V_{out} - V_{in}) / (V_{out}) = (20-10)/20 = 0.5 \Rightarrow 50\%$$

Inductor value:

$$L = \frac{V_{in}}{2 \Delta i_L D T}$$

Δi_L = Ripple current value in current = 10% of 5A = 0.5

$T = 1/f = 1/1\text{kHz}$

$$L = \frac{10}{2 * 0.5} * 0.5 * 0.001 = 5 \text{ mH}$$

Capacitor value:

$$C = \frac{V_{out}}{2 * R * \Delta V} D T$$

ΔV = ripple voltage in output voltage = 1% of 20V = 0.2 V

$R = 4 \text{ Ohms}$ (because V and I are fixed values, 20 V and 5 A)

$$C = \frac{20}{2 * 4 * 0.2} * 0.5 * 0.001 = 6.25 \text{ mF}$$

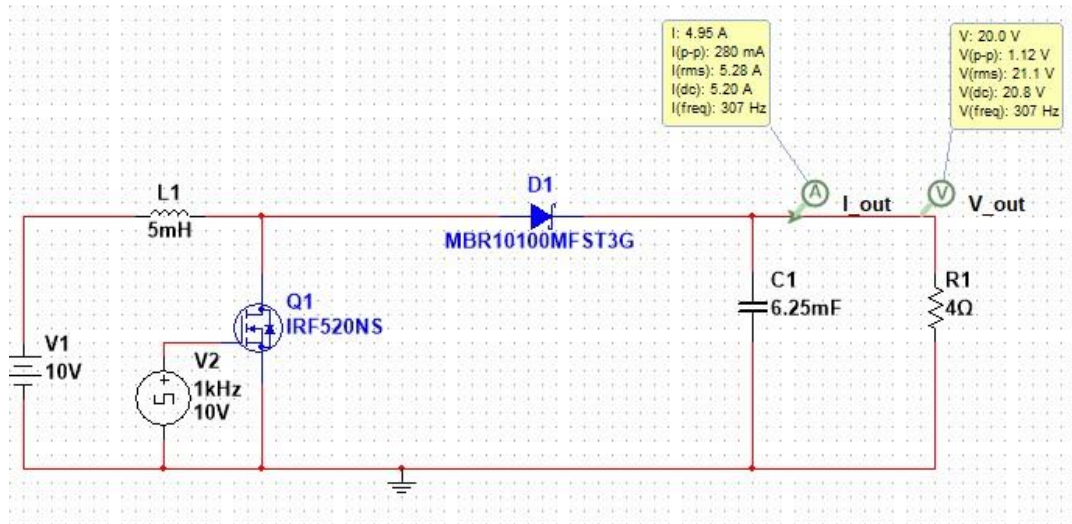


Figure 12 Running on interactive mode

$$I_{out} = 4.95 \text{ A}$$

$$V_{out} = 20.0 \text{ V}$$

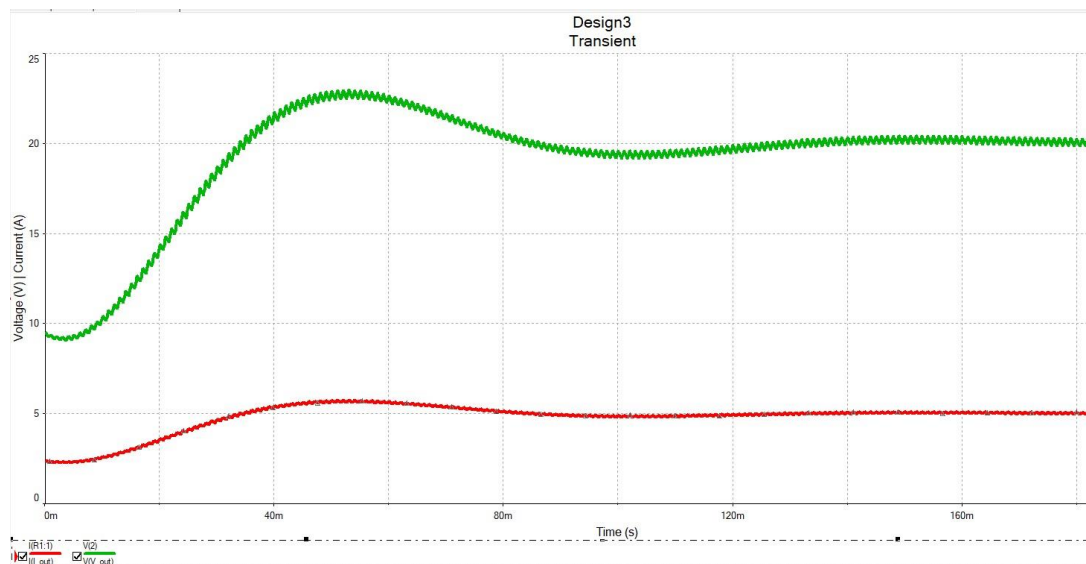
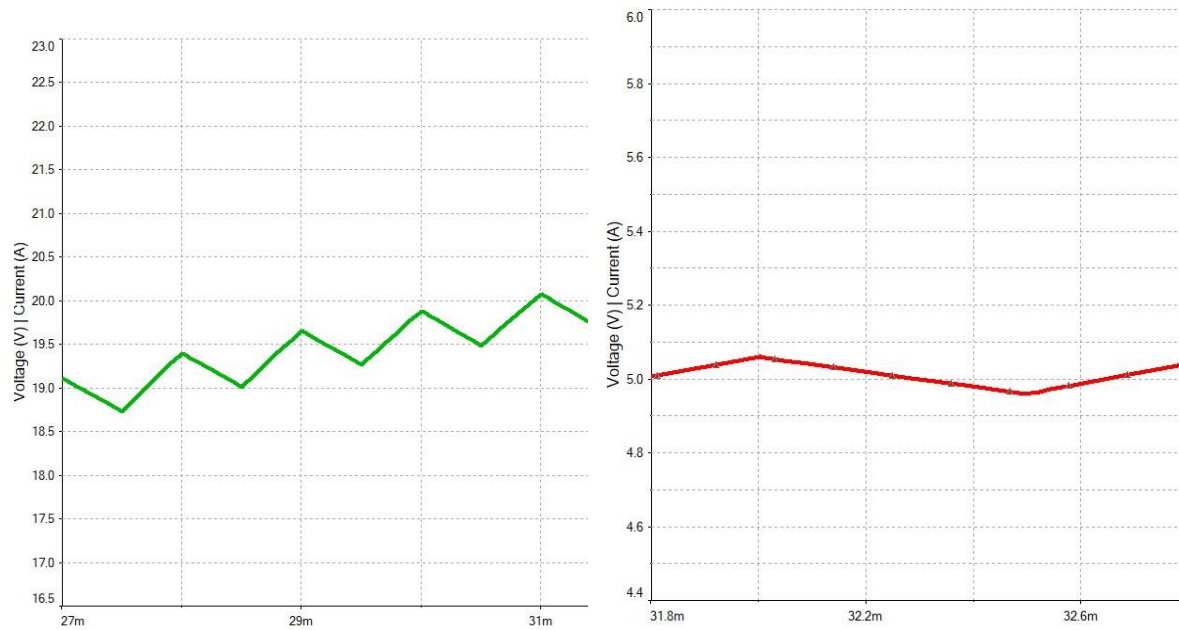


Figure 3 Graph of V_{out} (green) and I_{out} (red)



Ripples are triangular with edges, not smooth. Why?

Because Inductor is not directly attached with capacitor, so no LC filter (two pole filter) created at output side.

Single pole filter= Triangular ripples (Boost Converter)

Two pole filter= Smooth ripples (Buck Converter)

Boost converter applications

- Automotive applications
- Power amplifier applications
- Adaptive control applications
- Battery power systems
- Consumer Electronics
- Communication Applications

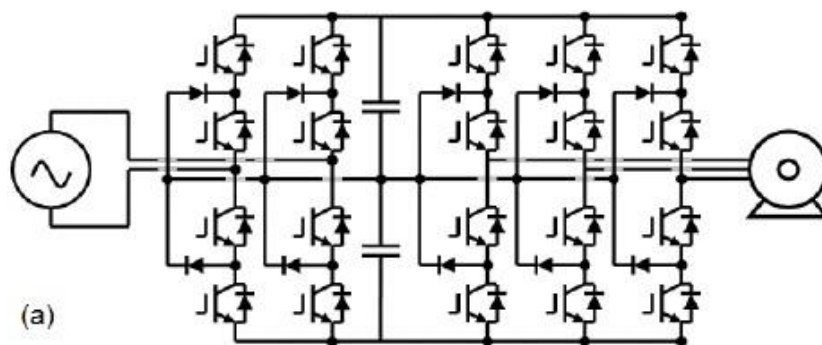


Figure High power, train drive system in Japan that uses boost converter

(b)

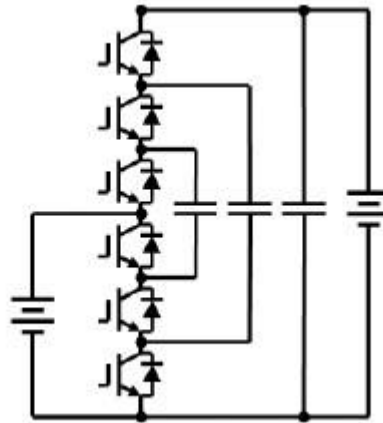


Figure Low power, DC-DC converter for automotive system

BUCK-BOOST CONVERTER

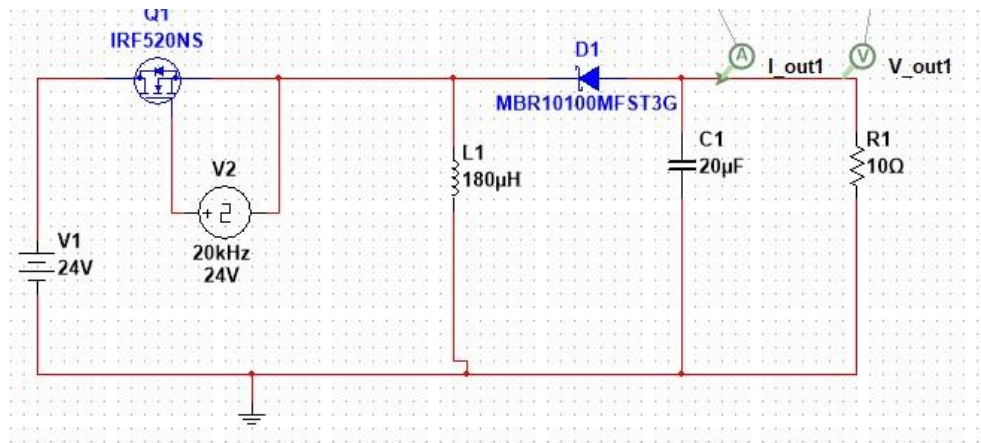
How does it work?

The main objective of a buck-boost converter is to receive an input DC voltage and output a different level of DC voltage, either lowering or boosting the voltage as required by the application. The design of a buck-boost converter is similar to a buck converter and boost converter, except that it is in a single circuit and it usually has an added control unit. The control unit senses the level of input voltage and takes appropriate action on the circuit based on that voltage.

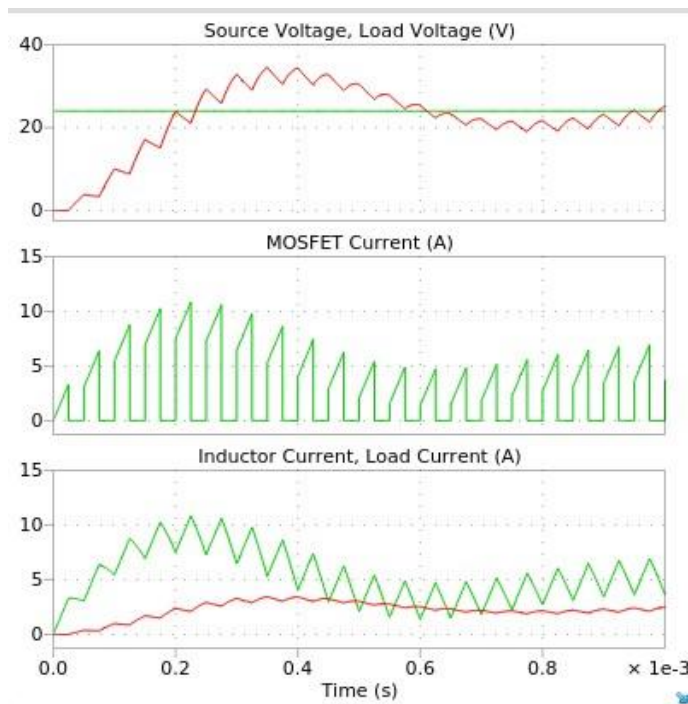
The circuit operation depends on the conduction state of the MOSFET:

- **On-state:** The current through the inductor increases and the diode is in blocking state.
- **Off-state:** Since the current through the inductor cannot abruptly change the diode must carry the current so it commutates and begins conducting. Energy is transferred from the inductor to the capacitor resulting in a decreasing inductor current and a voltage across the resistor with the opposite polarity compared to V_{in} . During steady state the circuit is said to operate:
 - in discontinuous conduction mode if the inductor current reaches zero
 - in continuous conduction mode if the inductor current never reaches zero

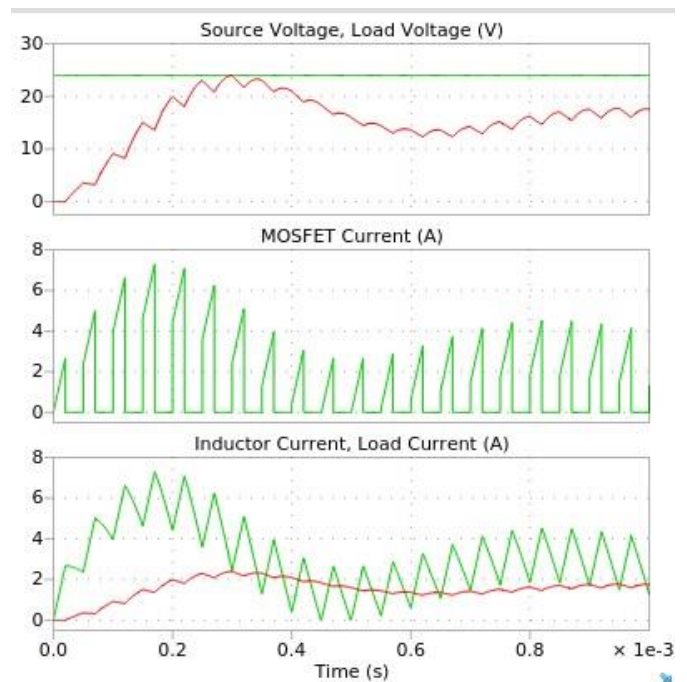
Buck-boost converter simulation



1.D=50%



2.D=40%

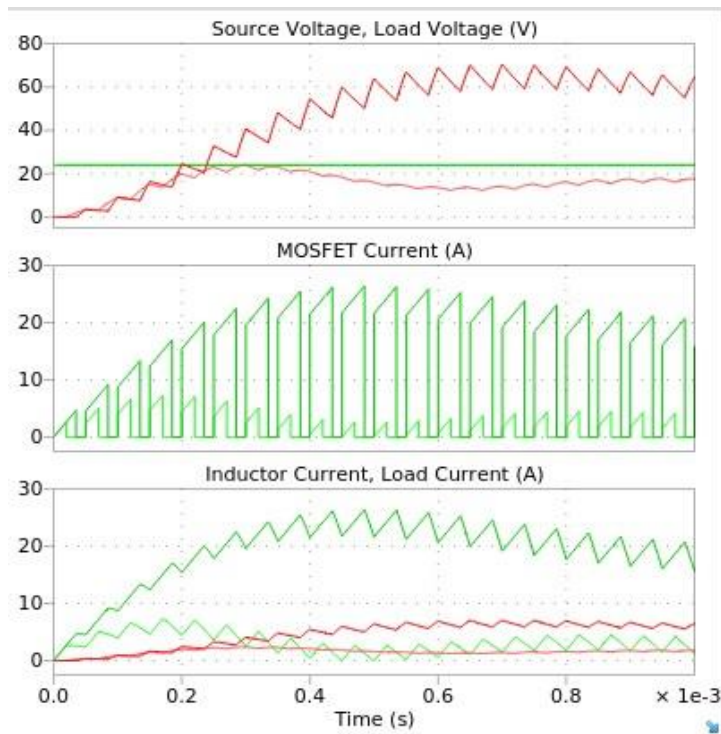


The circuit has two limits of operation. For a PWM duty cycle $D \rightarrow 0$ the output voltage equals zero, and for $D \rightarrow 1$ the output voltage grows toward negative infinity. In between those limits the output voltage in continuous conduction mode is given by:

$$V_{out} = -D/(1-D) \cdot V_{in}$$

(the graph simulations were taken from plexim.com simulator)

3.D=70%

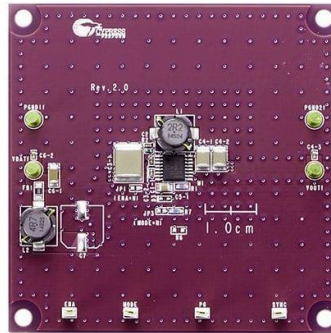


Buck-Boost applications

Buck boost converters make it easy to modulate voltage for a wide variety of popular applications including consumer electronics, power amplifiers, self-regulating power supplies and control applications.



This buck-boost DC/DC converter is both efficient and low-noise (up to 1.2MHz Ultralow noise PWM frequency). Its input voltage can range from 2.4V to 25V, while its output voltage can range from 1V to 25V, making it very versatile.



This 1-channel buck-boost DC/DC converter IC has four built-in switching FETs and can supply up to 2.4A of load current within an input voltage range from 2.5 V to 42 V. It Choose from a variety of output voltages (5.000V/5.050V/5.075V/5.100V/5.125V/5.150V/5.200V) and take advantage of a wide operating frequency range from 200 kHz to 2.1 MHz.

It is recommended for long-life battery-operated instruments, portable military radios, low power sensors and solar panel post-regulators/chargers.

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Buck-Boost Converter

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Buck DC/DC Converter

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Buck Boost Converter Application Notes

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Buck Converters and their cool applications

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Boost Converters-Working, Application and Advantages

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How does a Buck Boost Converter Work?

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